

Periodic Weight Measurement for Bedridden Patients Using a Pressurized Liquid-Filled Channel System Integrated with Hospital Beds

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Abstract— Variations in body fluids in a short time can lead to significant health issues. The risk due to body fluid variations increases for critically ill and bedridden patients. Changes in body weight in short intervals are the primary indicator of body fluid imbalances. The clinical practice for periodic monitoring of body weights of critically ill or bedridden patients includes a dedicated weighing scale or lift scale, detachable or built-in units of weighing scale on to the legs of hospital cots or as integrated bed scales attached permanently to the cot. However, these methods have limited mobility and usability in a practical setting and often require the involvement of multiple personnel to measure the body weight of a bedridden patient. Owing to such constraints, periodic monitoring of body weights is seldom recorded. In this work, we propose a novel measurement scheme for periodic monitoring of body weight. The prototype consists of an active (piston-based syringe pump) pressurized liquid-filled flexible and elastic channel sandwiched between the hospital cot and mattress. The required displacement of the syringe pump to achieve an optimal internal pressure is calibrated with the known weights. A calibration model ($R^2 = 0.99$, $p < 0.001$) is developed that maps the displacement of the piston pump in mm to known weights in kg for the range 0 to 90 kg. Using the calibration model, the accuracy and precision of the system are characterized at various loading conditions. A practical scenario of rising body weights was emulated as an invitro-experiment, resulting in a maximum error of 3%. The design of the system is modular, so it can easily be integrated with any type of hospital bed. This design is proof of concept for using alternative technologies for weight measurement over periodic intervals, where conventional designs seem cumbersome.

Keywords— weight measurement, weighing scale, body fluid imbalance, kidney failure, dialysis

I. INTRODUCTION

Body weight is a crucial indicator of an individual's overall health and holds significant clinical importance. In critically ill patients within Intensive Care Units (ICUs), monitoring

body weight and its variations is vital for assessing the efficiency of fluid management and the patient's health status. It is a direct indicator of the fluid balance effectiveness as well as the overall health of the patient. Approximately 60% of an individual's weight is due to body fluids [1]. Variations in fluid volume of 5-10% can lead to dehydration and edema [2], stemming from inadequate homeostasis maintenance, a situation often more acute in older adults [3]. The human renal system plays a key role in regulating fluid balance to preserve our 'dry' weight and prevent dehydration [4]. Individuals suffering from chronic kidney disease struggle to control fluid balance, potentially leading to edema unless managed, typically through dialysis, which may involve removing 2 - 20 kg of fluid to keep the dry weight stable [5]. Regular monitoring of weight is a method used to check for fluid imbalance [6], which helps in optimizing dialysis treatment [7], identifying issues with intravenous fluids (IV) therapy [8], and assessing nutritional health [9]. Changes in body weight are linked to poor homeostasis maintenance, causing health deterioration in the elderly [10]. Regular and precise weight measurements can significantly decrease health-related risks, especially in older and bedridden patients [11]. The periodicity of weight monitoring completely depends on the clinical applications.

While monitoring body weight is key to detecting fluid imbalances, accurately measuring the weight of bedridden or immobile patients in ICU settings poses significant challenges. Traditional methods require moving the patient onto a dedicated scale, a process that is not only impractical but can also be harmful or unethical for those in critical condition. Currently, the predominant technique for weighing immobile patients involves strain gauge-based load cell devices. These devices are either incorporated into lifting scales or roll-on scales, where the bed is moved onto the scale, or they involve detachable or built-in load cells on the bed's load-bearing structure. However, integrating load cells into hospital beds can significantly increase manufacturing costs, making such beds less accessible to many public health facilities. Detachable units, which need to be placed under each bed leg as needed, present practical difficulties and usability concerns in clinical environments. Moreover, the necessity to account for both the bed's structural weight and

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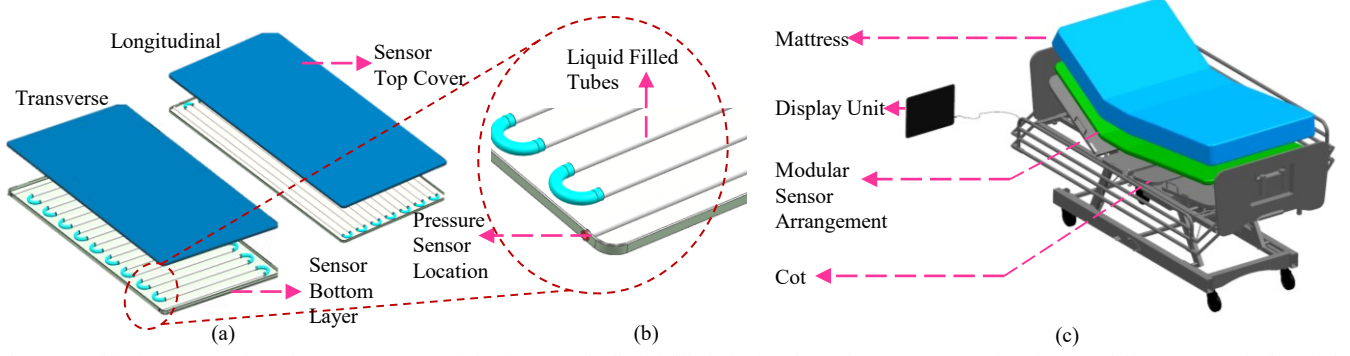


Fig. 1. Graphical representation of (a) arrangement of elastic tubes, (b) liquid filled elastic tube and pressure sensor location, (c) different parts of a hospital bed and modular sensor arrangement.

the patient's weight leads to an over-design of load cells for high accuracy at lower operational ranges, which escalates both the cost and size of these devices [12]. The risk of error propagation and failure in systems using multiple sensors, along with the potential for measurement errors or sensor damage, underscores the limitations of current methods. [13]. Therefore, there exists a need for an innovative, simple, and modular approach to facilitate accurate weight measurements for hospital beds. In our previous work [14], air-filled pillows and mattresses were explored for weight measurements; however, they had limitations.

In this work, we propose a sensing methodology and a measurement scheme to detect and record changes in the weight of bedridden patients. The sensing methodology needs to be part of the bed or mattress (just above or below the mattress) and not to be integrated into the cot (mechanical structure), as illustrated in Fig. 1. This will make the sensing modality flexible so that it can be incorporated into any hospital beds at minimum cost. Any add-on structural changes to the cot will attract higher production costs, making it unaffordable for public hospitals. The proposed system consist of a liquid-filled channels made of elastic and flexible tubes, pressurized to an set value actively by a piston-based syringe pump. The required displacement of the piston-based syringe pump to achieve the set internal pressure is calibrated with the known standard weights, and a calibration model is derived. The article further discusses the principle of measurement, the design of the system, and in-vitro experiments, followed by the results, discussions and limitations.

II. MATERIALS AND METHODS

A. Principle of Measurement

Any electro-mechanical measurement system typically consists of a forcing function expressed in terms of the process variable, a control function to attain a steady state and a damping function to ease out all the oscillations. In this work, a closed loop system (active system) is designed with feedback control on process variables. At constant temperature and for a given mass of liquid, there is an initial volume of liquid V_0 within the elastic tubes of defined dimensions. Due to the elastic nature of the tubes, the weight applied (W_p or W_v) on the system is transmitted to the fluid inside and results in a pressure change (ΔP). This phenomenon is approximated by the relation,

$$W_p = C_p \Delta P \quad (1)$$

where, C_p is the proportionality constant. In a passive system (without any feedback control), for a given volume of liquid

inside the tubes, the change in pressure due to the applied load can be measured, however, it is susceptible to the non-linearity of the system. In an active system, the internal fluid pressure is always kept constant by regulating the volume using a piston-based syringe pump and the amount of volume change (ΔV) required to keep the pressure constant for an applied load is measured (and calibrated), approximated by the relation,

$$W_v = C_v \Delta V \quad (2)$$

where, C_v is the proportionality constant.

Fig. 1 illustrates the modular arrangement of the sensor. The liquid-filled channels can be integrated in either longitudinal or transverse fashion between two support sheets, with the sensor located at the end of the channel. The whole arrangement can just be placed beneath the mattress without any structural change to the cot.

B. Objectives and Verification Strategies

The objective of this study includes,

1. To select the correct type of elastic and flexible tube that will transfer the pressure due to the weight applied to the liquid with minimal loss.
2. To characterize the optimal internal pressure for maximum sensitivity of the system (set value).
3. To construct a calibration curve and characterize the accuracy and repeatability of the system at the set internal pressure.
4. To emulate a realistic scenario of rising body weights and verify if the system can track the same.

C. System Design

The prototype consists of an elastic tube in the shape of "U" attached to a pressure sensor (Jupiter Electronics, India - 85CV-015G-8R, 0 – 15 psi, 0 – 100 mV) at the intersection. The tubing was initially primed with water as the liquid, ensuring no air bubbles or pockets were present in the system. Both ends of the U-tube were closed with ball valves or a combination of Non-Return valves (NRV) and solenoid valves. All the tubes (sensing area) were sandwiched between two thick acrylic sheet supports (180 cm x 70 cm x 1 cm), providing uniform loading and uniform volume change, as graphically represented in Fig. 2. The pressure sensor gives an output of 0 – 100 mV for the range 0 – 15 psi (0 – 775.72 mmHg), which was amplified by an instrumentation amplifier (Texas Instruments – INA125P). A precise excitation voltage of 10 V is maintained for the pressure transducer. The amplified voltage is acquired by the National Instruments (NI) 6216 DAQ card at a 1 kHz sampling rate. The pressure sensor was calibrated for reading in mmHg.

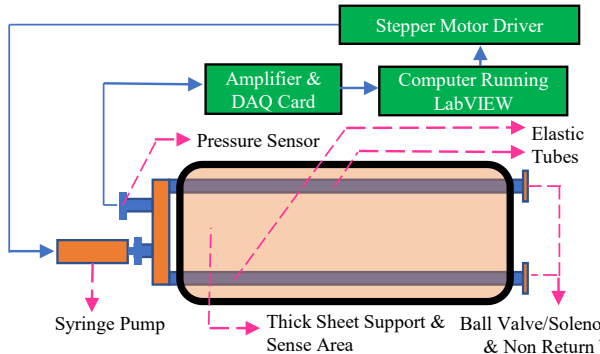


Fig. 2. Graphical representation of the system: hardware architecture

To implement a closed-loop control system, the process variable (pressure) within the elastic tubes must always be regulated by a control variable to reach a set value that gives higher sensitivity. The negative feedback system always tries to maintain optimal pressure. The optimal pressure was found experimentally. A piston-based syringe pump was connected at the base of elastic tubes laid in the shape of a U to regulate pressure. The custom-made syringe pump consists of a 150 ml syringe attached to a linear actuator (4 mm lead) controlled by a stepper motor (NEMA 17 - 42HM48-1684, 0.9° step angle) and its driver (Rhino Motion Control - RMCS 1106). The volume of the syringe pump was chosen such that the required change in pressure was within our operational range. The volume change of the syringe regulates the pressure to set value, realizing a closed loop negative feedback control system. The controller computes the linear displacement (in units of mm) proportional to the load acting on the tubes (and proportional to volume change) so that the syringe pump will displace the piston to maintain the optimal pressure. This linear displacement (control variable) is an estimate of the load acting on the tubes, and a calibration model between linear displacement and applied load will characterize the system. Closed-loop control will make the system more reliable and robust. It takes care of irregularities like control action over the output value and provides more flexibility in optimizing the system for its maximum performance. An overview of the control architecture is shown in Fig. 3.

D. Experiments on Selection of Tube

Selection of the correct type of elastic tube was made based on relative stiffness, step response of the pressure changes and approximate pressure loss (P_{loss}) at the media interface between the applied load and tube's surface. P_{loss} is calculated as,

$$P_{\text{loss}}(\%) = \frac{\left| P_m - \frac{W_{\text{ap}}}{A} \right|}{\frac{W_{\text{ap}}}{A}} \times 100 \quad (3)$$

where, P_m is the fluid pressure measured from the sensor attached to the tube, A is the approximate area of contact between the applied weight and tube, which will vary for different tubes and W_{ap} is the applied weight. This experiment is limited to a few variants of the commercially available flexible PVC pipe (T1, T2, T3) and silicon rubber tube (T4) of various thicknesses and stiffness as listed in TABLE I. Dynamic response of the pressure change was conducted by applying standard weights on the tubes in the passive type of system.

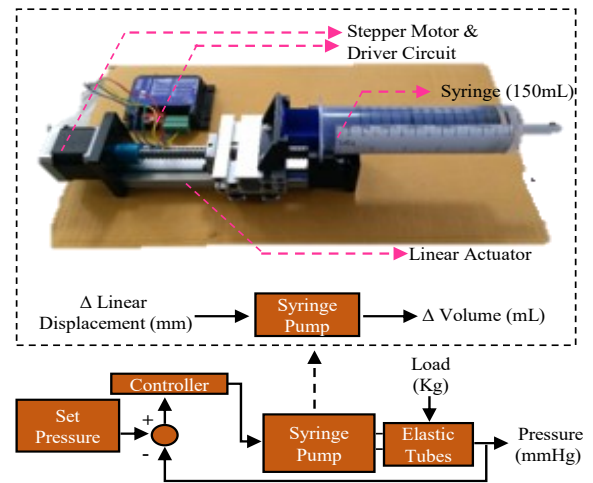


Fig. 3. Control architecture of the system with parts of the syringe pump highlighted.

E. Experiments on Sensitivity Characterisation

The syringe pump can vary the internal pressure based on the volume injected or ejected from the system. Experiments were conducted by varying the pressure from 50 mmHg to 300 mmHg, in order to identify an optimal pressure which can be used as the set pressure of maximum sensitivity. The syringe pump is initially kept at the extended position with minimum pressure applied on the fluid by the piston. The syringe pump is then allowed to compress in steps of 50 mmHg up to 300 mmHg. For every 50-mmHg change, an external load of 15 kg is applied (and then removed) on the sensing area, for which the change in pressure is noted. All readings were obtained for the steady state. Hence, adequate settling time must be provided when the syringe pump is compressed for every 50 mmHg, when the load is applied and when the load is removed so that the oscillations created within the liquid will dampen out.

F. Calibration Curve

A calibration model was made based on several sets of data collected between the linear displacement (mm) of the syringe pump and the applied load (0 – 90 kg). The linear displacement of the syringe pump is obtained from the angular rotations of the stepper motor and then converted based on the pitch and lead of the linear actuator used. Error analysis and repeatability studies were conducted for weights varying from 0 – 90 kg in units of 15 kg based on the obtained calibration model. All readings were obtained for the steady state.

G. Emulating a Practical Scenario: In-vitro Experiment

A practical scenario was emulated, and the objective was to track small changes in weight over and above a person's dry weight and perform an error analysis based on the calibration model obtained previously. The experiment involves loading & unloading small units of weights over and above the dry weight of 61 kg. The participant was asked to lie on the bed in the supine position, and smaller value weights in units of 3 kg, 2 kg, 1 kg and 0.5 kg were loaded to emulate a real-time change in weight, typical of a patient suffering from fluid accumulation. Weight placement follows a uniform distribution. All readings were obtained for the steady state.

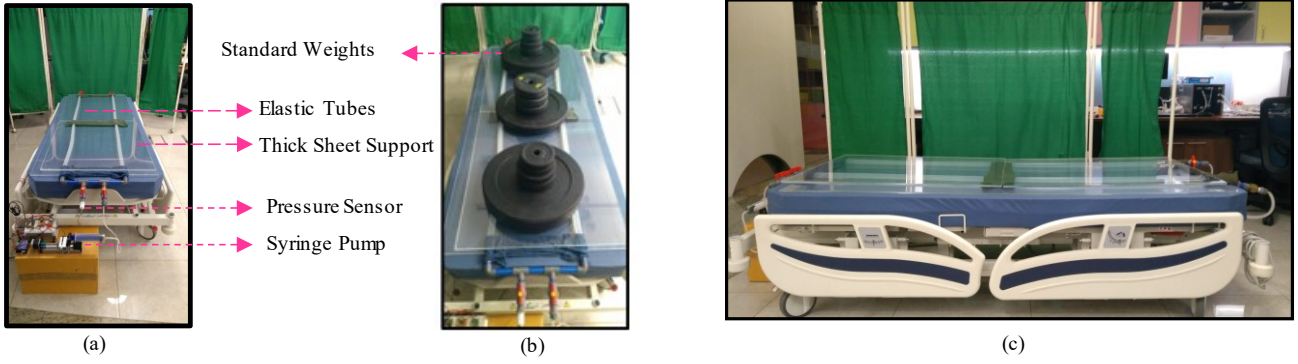


Fig. 4. (a) Active system in unloaded condition. (b) active system in loaded condition, (c) side view of the bed and sensor module arrangement

H. Statistical Analysis

All the group average measurements were reported as mean $\pm$ standard deviation. The variability in measurements was analyzed using the coefficient of variation (CoV). The CoV was obtained as the ratio of respective standard deviation to mean, expressed as a percentage. A polynomial regression analysis (curve fitting) was used to derive the linear calibration model. The linear regression analysis and correlation coefficient were used to determine the association and correlation between variables. The level of significance of $\alpha = 0.05$ was used for all tests. A p-value < 0.05 was used as a statistical significance to reject the associated null hypothesis.

III. RESULTS AND DISCUSSION

A. Selection of Tube

A heuristic approach was adopted to select the type of elastic tubes. A silicon rubber-based tube (T4) with outer diameter (OD) of 25 mm and internal diameter (ID) of 19 mm was selected, based on low-pressure loss. Investigating parameters are summarized in Table I. The pressure loss was attributed to the stiffness of the tube. The more elastic content, the less the pressure-transmission loss. The dynamic response depends on the tube's rigidity, thickness and material. A stiffer tube will give a faster response at the cost of a high-pressure transmission loss.

B. Sensitivity of the system

The effect of sensitivity on initial pressure is illustrated in Fig. 5. From the plot, 50 mmHg is identified as the optimum internal pressure for maximum sensitivity of 1.7 mmHg/kg, which is translated to 0.23 mm/kg or 20.7°/kg for the syringe pump. Any pressure below 50 mmHg did not provide any appreciable change in sensitivity. The following parameters were identified that influence the sensitivity of the system: elasticity of the tube, the contact area of the acrylic support sheet on tubes (defined as the sensing area), the initial pressure of the system and the set pressure in an active system, sensitivity & linearity of the pressure sensor.

C. Calibration Curve and Reliability of the System

The calibration model obtained between linear displacement of the syringe pump and applied load is parabolic in nature with $R^2 = 0.993$ over the entire range of measurement,

illustrating that the sensitivity of the system is not constant, as shown in Fig. 6(a). Sensitivity can be assumed constant for smaller intervals of range. A repeatability study gave less than

TABLE I TUBE SELECTION

Specifications	Investigating Parameters			
	T1	T2	T3	T4
Length (mm)	320	320	320	320
Outer Dia.(mm)	21.38	11.45	21.38	25
Thickness (mm)	1.39	1.32	3	6
Relative siffness	medium	high	medium	low
Rise Time (sec)	59	2	60	45
Set. Time (sec)	122	22	137	120
P_{loss}	48%	97%	44%	25%

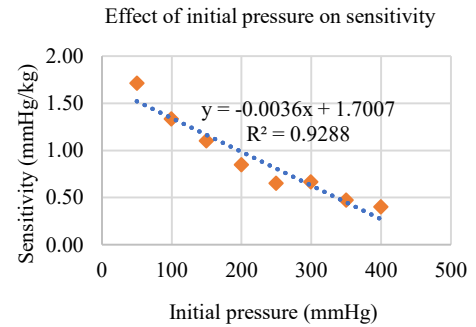


Fig. 5. Effect of initial internal fluid pressure on sensitivity of the system.

4% coefficient of variance (CoV) for applied loads in the range 45 – 90 kg. The minimum error was observed for the loads in the same interval with a maximum value of less than 3%. Hence, an optimal operational range is limited to 45 – 90 kg to have the least errors. Over the entire range, the maximum error was observed to be as high as 18% and repeatability at 11%. The capacity of this system was limited to 90 kg due to the limit imposed by the volume of the syringe pump. The agreement between the measured and applied load is shown as a linear fit with $R^2 = 0.9976$, as illustrated in Fig. 6(b).

D. Emulating a Practical Scenario

The changes in weight obtained from the calibration model followed a linear trendline with $R^2 = 0.9976$, as shown in Fig. 6(e). The associated error was less than 3%, as shown in Fig. 6(f). The resolution obtained was 500 g. This experiment was conducted to validate the calibration curve over a range of weights 60 – 75 kg, where the errors observed were minimal, and to understand the practical challenges when a human

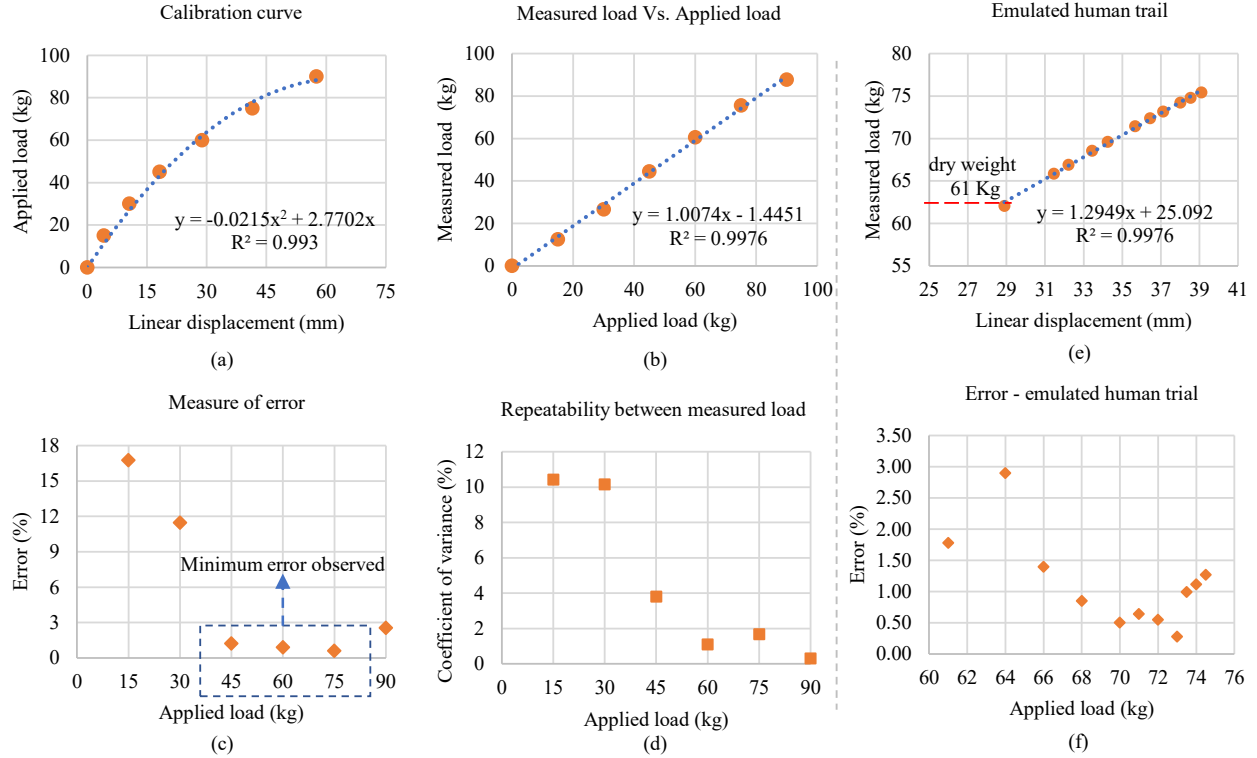


Fig. 6. (a) Calibration curve, (b) verification of the calibration curve with actual loading, (c) error analysis for the range 0 kg - 90 kg loading, (d) repeatability measure for the range 0 kg - 90 kg loading, (e) emulated in-vivo experiment, (f) error analysis in the in-vivo experiment

participant is acting as the base weight or dry weight instead of the standard weights.

E. Limitations and Future Scope

Although the system was able to track the case of a practical scenario of rising body weights with a resolution of 500g, which might suffice for a range of clinical use cases. The calibration model obtained was nonlinear. It is desirable to have a linear system to improve accuracy. The current calibration model is not valid for non-uniform load distribution and is currently valid only for supine position. The liquid filled in the system can be changed to silicone oil for improved settling time. The capacity is limited to 90 kg due to the limited volume of the syringe pump. Motion artefacts due to human subjects are only manually filtered as false positives. A 200 g resolution was observed when the model was scaled down by a factor of 75%, as the sensitivity was increased by having a lower area of contact between tubes and the support sheet. Future works will include implementing multiple smaller units of the measuring system to be kept beneath the mattress to improve sensitivity, capacity and linearity. The current version of the prototype does not differentiate motion artifacts and weight changes. Development of intelligent algorithms are in progress to identify and remove the effects of motion artefacts. The proposed system is also under investigation to be used as a balistocardiogram platform, to pick the mechanical vibrations of the heart and use deep learning frameworks [15], [16] [17], [18] for heart rate and its variability detection. Integrating this system with ultrasound-based measurement modalities [19], [20], [21], [22], [23], [24] for assessing cardiovascular health are also in progress.

IV. CONCLUSION

In this work, we have presented an exploratory sensing module which utilizes the pressure change within liquid-filled elastic tubes kept beneath a hospital bed mattress for periodic weight monitoring. Changes in body weight in short intervals are the primary indicator of body fluid imbalances. A calibration model is made with $R^2 = 0.993$ for a range of 0 – 90 kg, and the system was able to track the smaller changes in body weight emulated in a human trial experiment with an error of less than 3% for a resolution of 500 g. The CoV for operational ranges of 45 – 90 kg was observed to be less than 4%. The error for the same operational range was observed to be less than 3%. The maximum capacity of the measure was 90 kg. The system proved to be a feasible approach, and the sensing module can be easily coupled to any hospital bed.

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